北京郵電大學

Queen Mary University of London

For examiners' use only

1	
2	
3	
4	
Total	

EBU5212 A

Joint Programme Examinations 2023/24

EBU5212 Communication and Networks

Paper A

Time allowed 2 hours

Answer ALL questions

Complete the information below about yourself very carefully.

QM student number										
BUPT student number										
Class number										

NOT allowed: Any Electronic devices including electronic calculators and electronic dictionaries.

INSTRUCTIONS

1. **You must NOT take answer books, used or unused, from the examination room.**
2. Write only with a black or blue pen **and in English**.
3. Do all rough work in the answer book – **do not tear out any pages**.
4. If you use Supplementary Answer Books, tie them to the end of this book.
5. Write clearly and legibly.
6. **Read the instructions on the inside cover.**

Examiners

Dr Yan SUN, Dr Habiba Akter

Copyright © Beijing University of Posts and Telecommunications & © Queen Mary University of London 2023

Instructions

Before the start of the examination

1) Place your BUPT and QM student cards on the corner of your desk so that your picture is visible.

2) Put all bags, coats and other belongings at the back/front of the room. All small items in your pockets, including wallets, mobile phones and other electronic devices must be **placed in your bag in advance**. **Possession of mobile phones, electronic devices and unauthorised materials is an offence.**

3) Please ensure your mobile phone is switched off and that no alarm will sound during the exam. **A mobile phone causing a disruption is also an assessment offence.**

4) Do not turn over your question paper or begin writing until told to do.

During the examination

1) You must not communicate with or copy from another student.

2) If you require any assistance or wish to leave the examination room for any reason, please raise your hand to attract the attention of the invigilator.

3) If you finish the examination early you may leave, but not in the first 30 minutes or the last 10 minutes.

4) For 2 hour examinations you may **not** leave temporarily.

5) For examinations longer than 2 hours you **may** leave temporarily but not in the first 2 hours or the last 30 minutes.

At the end of the examination

1) You must stop writing immediately – **if you continue writing after being told to stop, that is an assessment offence.**

2) Remain in your seat until you are told you may leave.

Question 1

a) **Figure 1** shows a form of Huffman tree. Determine the code efficiency of this Huffman tree.

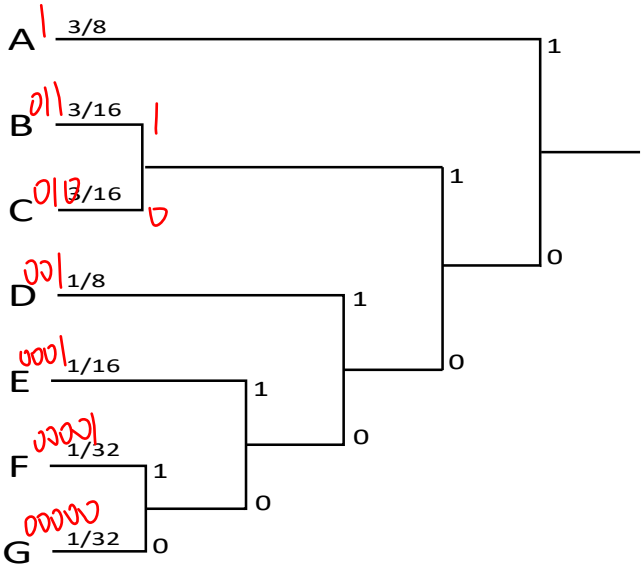


Figure 1

[12 marks]

	Do not write in this column
A B C D E F G	
probability 3/8 3/16 3/16 1/8 1/16 1/32 1/32	
code 1 011 010 001 0001 00001 00000	
code size 1 3 3 3 4 5 5	
$z_i \cdot \log_2 \frac{1}{p_i}$ 1.4 2.4 2.4 3 4 5 5	
$H = \sum p_i z_i = 2.3625$ bits/symbol	
$L = \sum p_i L_i = 2.4375$ bit/symbol	
$\eta_{code} = \frac{H}{L} = 96.9\%$	

b) A systematic linear block code with the following parity check matrix H is employed to protect information against errors:

$$H = \left[\begin{array}{ccc|cccc} 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 1 \end{array}\right]$$

[13 marks]

i) Find all the codewords via G matrix (generator matrix).

(8 marks)

ii) Decode the following received sequence r = 1111010.

(5 marks)

	Do not write in this column

Page **5** of **20**

$H^T = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$Y = 1111010$

$YH^T = [1001]$

$eH^T = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

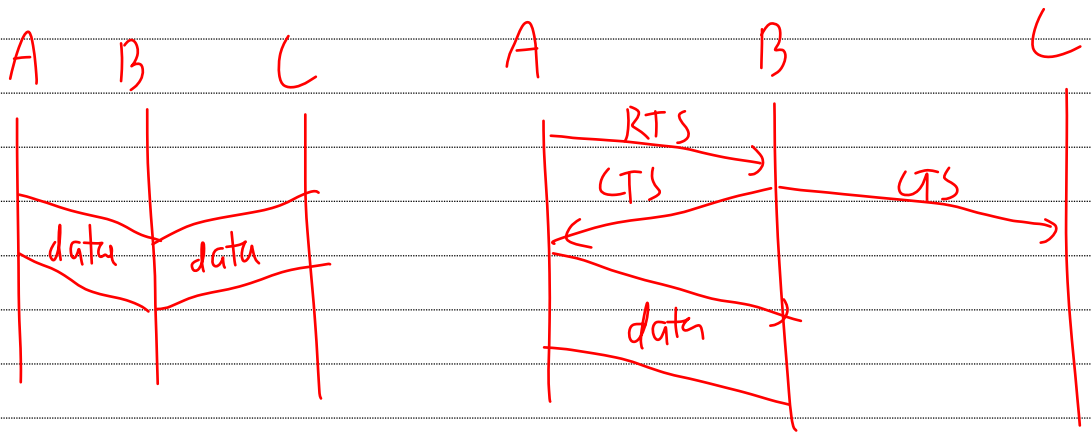
	13 marks

Question marking: $\frac{}{12} + \frac{}{13} = \frac{}{25}$

Question 2

a) What is the "hidden terminal" problem? Provide your answer with the aid of illustration.

[7 marks]

	Do not write in this column
 Assume there are 3 nodes A, B, C, A is B's neighbor, B is C' neighbor (A & C don't know each other). When A send data to B, C can't detect that A is sending to B and also sent data to B at same time. so there exist a collision. Solution: Add a CTS frame before sending data. After A send RTS frame, B should broadcast CTS frame to its neighbors When A receive CTS, it know B is waiting for data and start to send When other neighbor(C) receive CTS, it know B is waiting other node to send data C will stop to send data until there is a time-out and no CTS.	
	7 marks

b) Describe how host computer 192.168.1.2 is able to deliver Internet Protocol datagrams to host 192.168.1.4 on the same Ethernet network, via bridge BR, illustrating your answer with reference to **Figure 2a**. Your answer should include a full description of the role played by the Address Resolution Protocol (ARP) and assume that no previous communication has taken place for some time.

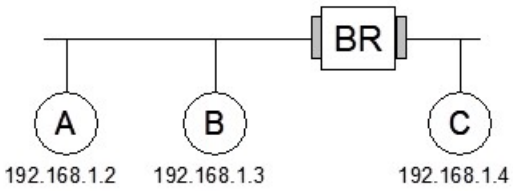


Figure 2a

[10 marks]

	Do not write in this column
ARP role: address resolution protocol is translate IP address into MAC address	
A先check自己的cache，无C's MAC	
how to deliver: ① ARP request: A request C的MAC address	
② ARP response: C response A and tell it MAC address	
when A know C's MAC address, 存入自己的cache	
③ A create IP datagram, destination = 192.168.1.4	
④ A encapsulate IP datagram into Ethernet frame. It's destination MAC address is C' MAC address	
⑤ Bridge BR收到Ethernet frame后check destination MAC地址 并forward it to port that connect most C.	
A B C 通过BR连在同一Ethernet下	
	10 marks

c) The bridge is now replaced with a router RT, as shown in **Figure 2b**. Explain any differences that would arise in the process of how host 192.168.1.2 delivers data to host 192.168.2.4. Once again, assume that no previous communication has taken place for some time and the subnet mask is /24.

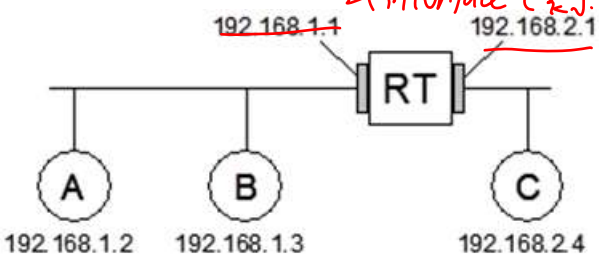


Figure 2b

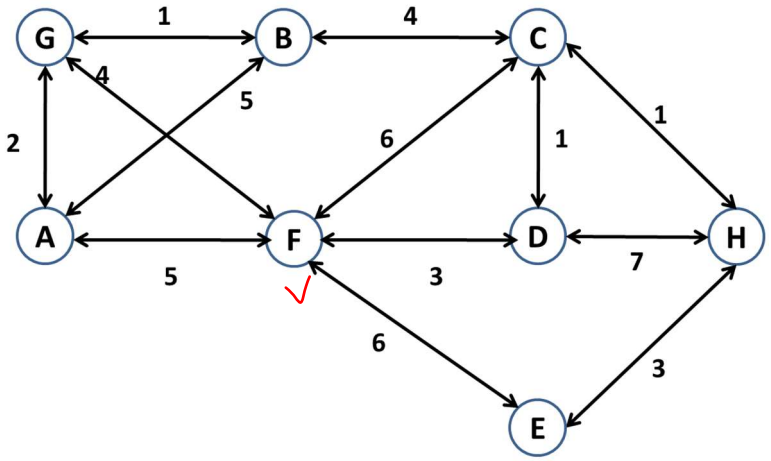
[8 marks]

	Do not write in this column
A、C处于同subnet 流程：A在subnet上广播ARP request，RT收到后 向A发ARP响应（包含RT 192.168.1.1接口的MAC地址） A收到后将192.168.1.1的IP地址和MAC地址存入ARP cache A create IP data 目标192.168.2.4 A将其放入Ethernet frame 目标采从RT获得192.168.1.1的MAC RT收到后check IP data的目标IP(192.168.2.4) RT根据routing table发现目标IP在另一子网将其转发入192.168.2.1接口 RT向192.168.2.1接口发ARP请求，C收到后回ARP响应，告诉自己的MAC RT有MAC，将IP data封装发送 不同：Bridge (BR) 工作在data link layer，它通过MAC地址转发frame 在同一Ethernet下可转发frame到目标port Router (RT) 工作在Network Layer，根据IP地址路由，A要发不同子网时 有IP路由决策和多个ARP request和data encapsulation	8 marks

Question marking: $\frac{}{7} + \frac{}{10} + \frac{}{8} = \frac{}{25}$

Question 3

a) Describe how would you determine the least-cost tree for the topology below from node D using Dijkstra's algorithm. You should also illustrate the final tree topology.



[12 marks]

	Do not write in this column
candidate wt Tree	
$R_D, R_D, 0$	
$R_D, R_L, 1$ 1 $R_D, R_D, 0$	
$R_D, R_F, 3$ 3	
$R_D, R_H, 7$ 7	
$R_D, R_F, 3$ 3 $R_D, R_D, 0$	
$R_L, R_H, 1$ 2 $R_D, R_L, 1$	
$R_L, R_B, 4$ 5	
$R_D, R_F, 3$ 3 $R_D, R_D, 0$	
$R_L, R_B, 4$ 5 $R_D, R_L, 1$	
$R_H, R_E, 3$ 5 $R_L, R_H, 1$	
$R_L, R_B, 4$ 5 $R_D, R_D, 0$	
$R_H, R_E, 3$ 5 $R_D, R_L, 1$	
$R_F, R_A, 5$ 8 $R_L, R_H, 1$	
$R_F, R_G, 4$ 7 $R_D, R_F, 3$	
$R_H, R_E, 3$ 5 $R_D, R_D, 0$	
$R_F, R_A, 5$ 8 $R_D, R_L, 1$	
$R_B, R_G, 1$ 6 $R_L, R_H, 1$	
	$R_D, R_F, 3$
	$R_L, R_B, 4$

$R_F, R_A, 5$ 8 $R_D, R_D, 0$

$R_B, R_G, 1$ 6 $R_D, R_C, 1$

$R_C, R_H, 1$

$R_D, R_F, 3$

$R_C, R_B, 4$

$R_H, R_E, 3$

$R_D, R_D, 0$

$R_D, R_C, 1$

$R_C, R_H, 1$

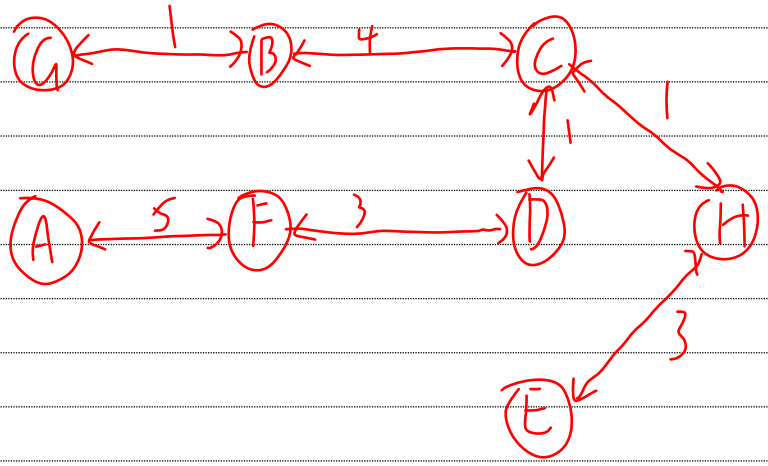
$R_D, R_F, 3$

$R_C, R_B, 4$

$R_H, R_E, 3$

$R_F, R_A, 5$

$R_B, R_G, 1$



12 marks

b) Applying the Distance Vector Algorithms to the topology in a), produce the routing table at node F.

[7 marks]

Do not write in this column

table 1 F's neighbor cost

node	cost
A	5
G	4
C	6
D	3
E	6

table 2 routing table that neighbor send to F

to / from	A	G	C	D	E
A	0	2	7	8	11
B	3	1	4	5	8
C	7	5	0	1	4
D	8	6	1	0	5
E	11	9	4	6	0
F	5	4	4	3	6
G	2	0	5	6	9
H	8	6	1	5	3

Then calculate the cost from F to other nodes

Table 3 : F's routing table

Dest	Distance	First hop
A	5	A
B	5	G
C	4	D
D	3	D
E	6	E
F	0	-
G	4	G
H	7	C

7 marks

c) Produce the updated routing table at node F with Distance Vector algorithm when node D is disconnected with node F and node G moves away from the topology in b).

[6 marks]

Do not write in this column

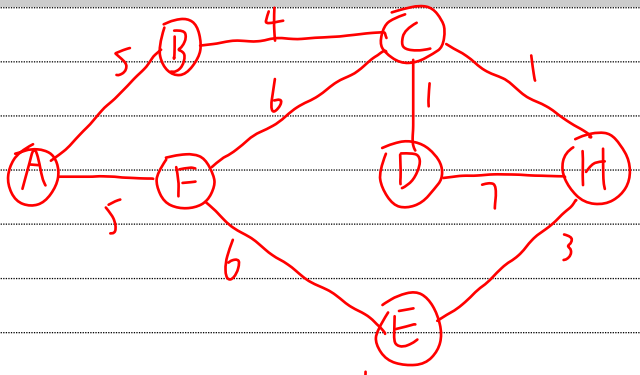


table 1 F's neighbor cost

neighbor	cost
A	5
C	6
E	6

table 2 routing tables from neighbors to F

to \ from	A	C	E
A	0	9	11
B	5	4	8
C	9	0	4
D	10	1	5
E	11	4	0
F	5	6	6
H	10	1	3

Then calculate the cost from F to other nodes

F's routing table

Destination	cost	First hop
A	5	A
B	10	A
C	6	C
D	7	C
E	6	E
F	10	A
H	7	C

6 marks

Question marking: $\frac{\quad}{12} + \frac{\quad}{7} + \frac{\quad}{6} = \frac{\quad}{25}$

Question 4

a) Considering the traditional TCP in wired IP network,

[14 marks]

i) What are the advantages and disadvantages of the fact that TCP acknowledgements are cumulative?

累积确认

(9 marks)

ii) What can be deduced from the reception of duplicate acknowledgements?

(2 marks)

iii) How is this information used in TCP Reno to maintain channel utilisation?

(3 marks)

	Do not write in this column
i) advantage:	
① easy to generate: receiver just send 1 TCP cumulative ACK and can tell transmiter that they have received all the data before the SN. Don't need to send ACK for each data.	
② tolerance ACK-loss: 即使有ACK丢失, 后续只要有ACK给transmitter 仍可知 which data haven't been received. 不需retransmit 所有 data	
disadvantage:	
① transmiter 只知 last contiguous number of data, 不知 具体哪个data接收	
② 有data丢失时 transmiter 可能重传多个 segment, 不仅只传丢失那个	
③ 难以实时调整 window size.	

2) ① Receiver haven't received所有data, 出现segment loss

② 可能有congestion

3) ACK对channel利用率影响

① 调整cwnd: 当transmiter发的data被receiver全接收并发ACK确认, TCP Reno会按congestion control算法增加cwnd

② 判断congestion: 当transmiter 3-duplicate ACKs, 说明出现congestion, faithly retransmit segment且减小cwnd, 保证信道利用率

③ 根据ACK中window size调整transmiter window size

14 marks

b) TCP congestion control uses 4 basic mechanisms known as Additive Increase (AI), Multiplicative Decrease (MD), Slow Start and Fast Recovery. Briefly explain each of those 4 basic mechanisms with figure illustration.

[11 marks]

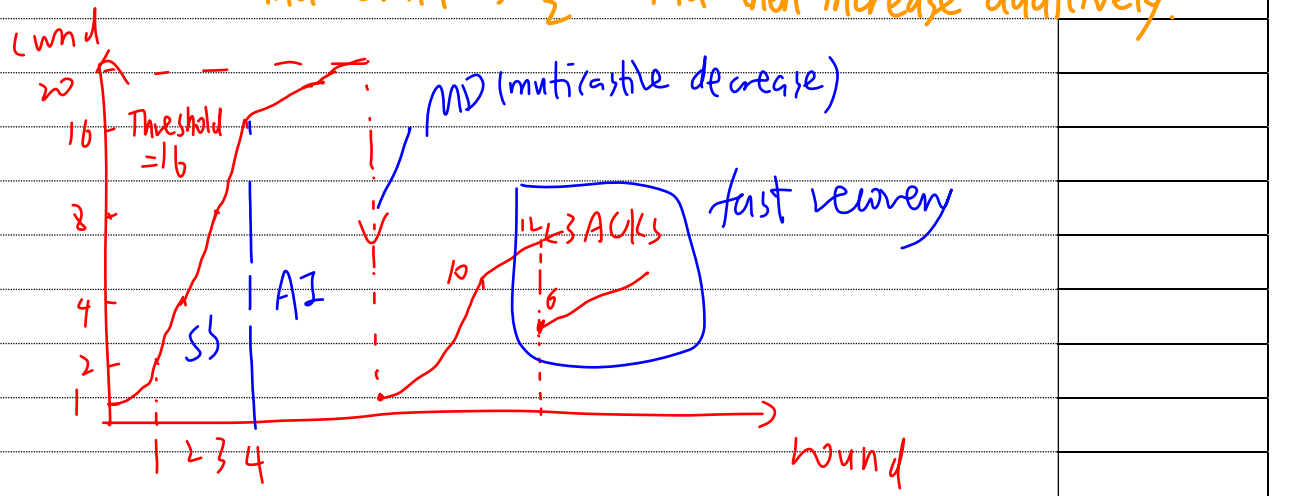
Do not write in this column

AI: cwnd reach threshold 后 run congestion avoidance algorithm
then cwnd size will additively increase

MD: If cwnd reach time-out or transmiter get 3-ACKs
说明出现congestion, cwnd 将 multiplicatively decrease
若是 time-out cwnd → 1
3-ACKs cwnd → $\frac{cwnd}{2}$

SS: when cwnd < threshold, cwnd will increase exponentially
until cwnd = threshold

Fast recovery: when transmitter get 3 duplicate Ack (3-ACKs condition)
快重传 transmitter will retransmit corresponding segment at once
and cwnd → $\frac{cwnd}{2}$ and then increase additively.



Do not write in this column

2023-2024

Rough Working

Page 18 of 20

Do not write in this column

2023-2024 Rough Working

Page 19 of 20

Do not write in this column

2023-2024

Rough Working

Page 20 of 20